Indicative solution of the De/Radicalisation I case (EN)

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September 2012

The de-radicalization case allows to explore how/why activism may become more extremist/harmful or moderate/harmless. This case is moderately difficult because (i) it is not staged, (ii) it is somewhat explorative, (iii) the effects of policies are rather counter-intuitive, and (iv) it is difficult/impossible to make a highly aggregated CLD of this model.

Students need to model this issue (see Figure 1), test the model, simulate the model over a time horizon of 100 years starting in 1980 and make graphs of key variables (Figure 2(b) and 2(c)).

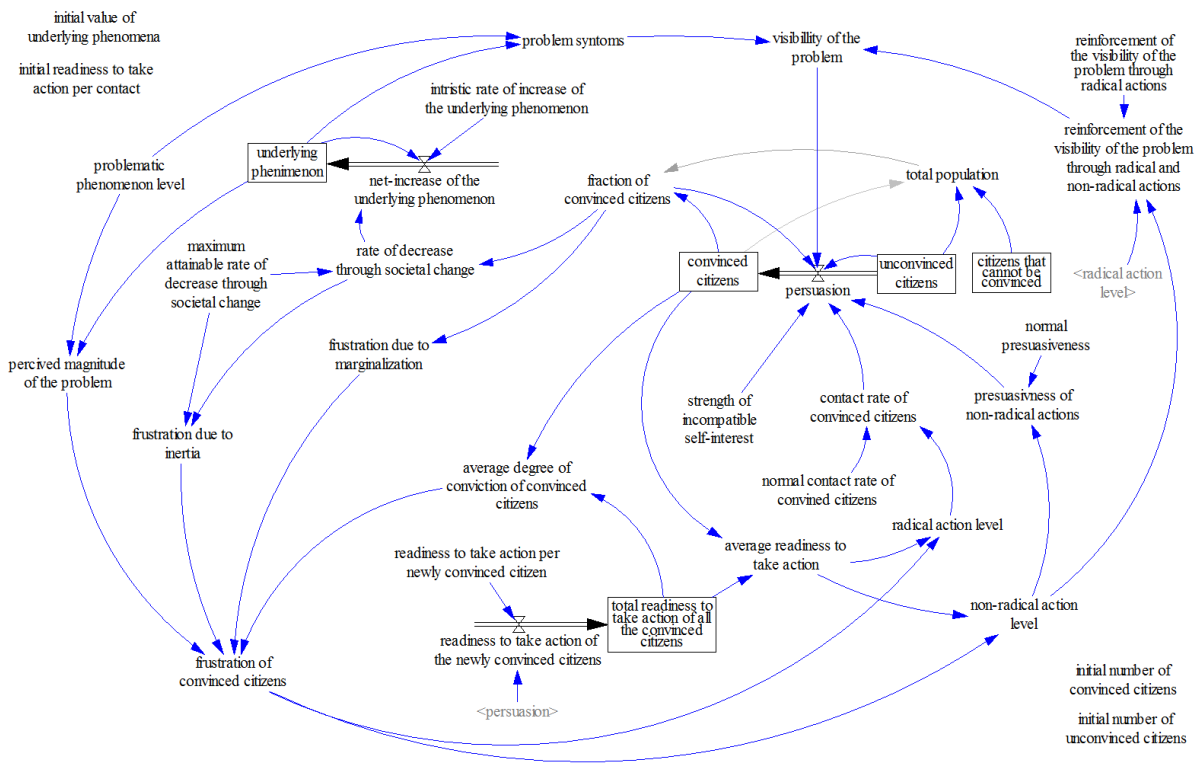
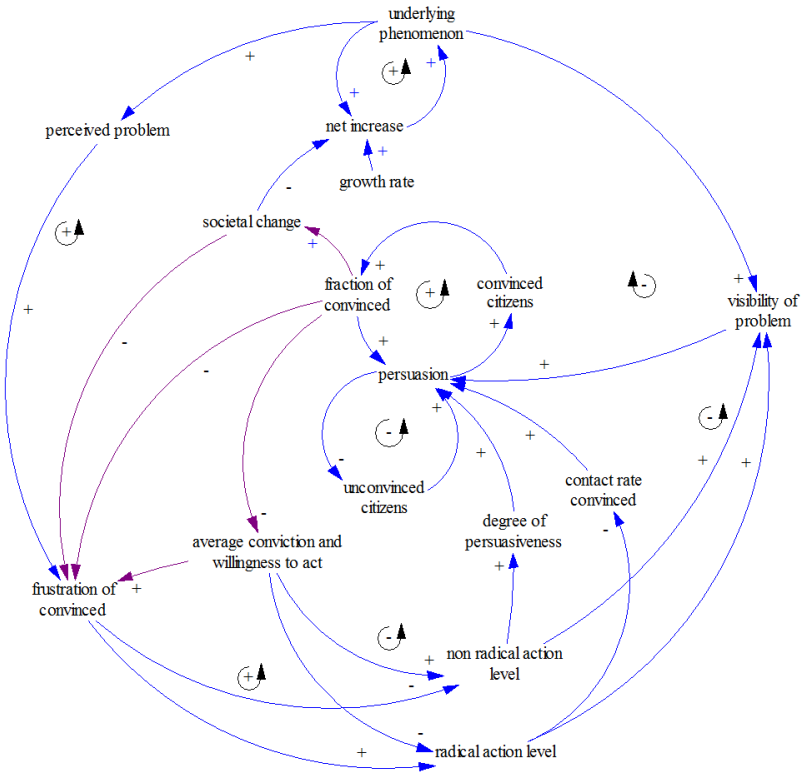


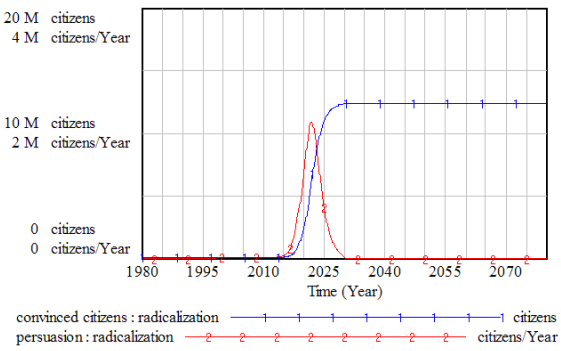
Figure 1: Full Stock-Flow Diagram of the de/radicalisation model

Then they need to test the sensitivity/uncertainty of the model for changes in several parameters (Figure 2(d)) and draw conclusions: significant changes in five out of six parameters lead to a fundamentally different mode of behavior (see Figure 2(d) – the blue curve represents the base case mode of behavior).

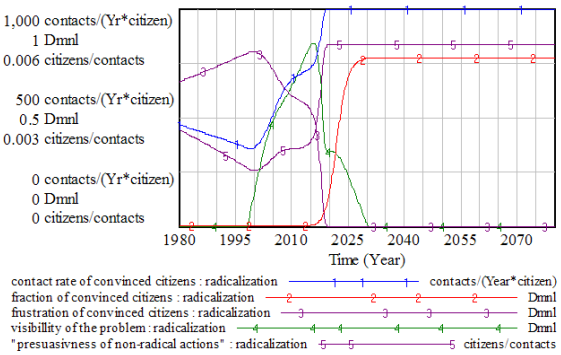
Students are asked to make two interesting and consistent exploratory scenarios based on the results of this sensitivity analysis: the very different scenarios displayed in Figure 2(e) can be distilled since this model is characterized by a strong bifurcation. Students also need to make



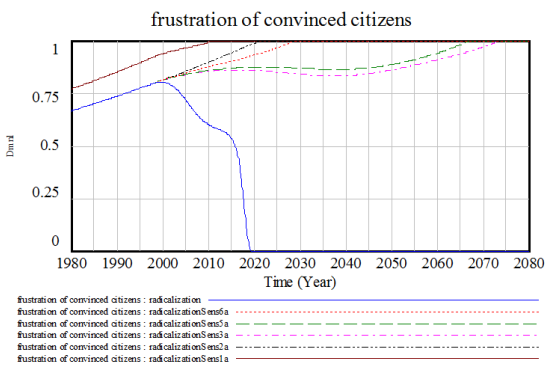
(a) CLD of the de/radicalization model



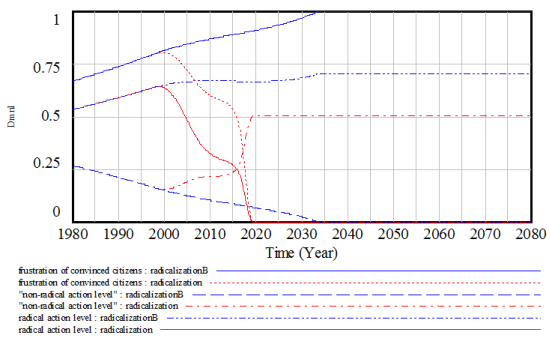
(b) Evolution of the stock of *convinced citizens* (blue –1–) and *persuasion* flow (red –2–) in case of deradicalization



(c) Evolution of main drivers of *persuasion* flow in case of deradicalization



(d) Behavior mode sensitivity (*frustration of convinced citizens*) for 5 out of 6 parameters



(e) Two distinct scenarios: deradicalization (red) versus further radicalization (blue)

Figure 2: CLD of the de/radicalization model (a), and ***de*radicalization** mode (b & c) versus **radicalization** mode (d & e) of the de/radicalization model

a CLD of this simulation model, and use it to explain the link between structure and behavior. However, this question may be rather hard because it is difficult to make an simple/aggregated CLD of this model – a rather detailed CLD is displayed in Figure 2(a).

Finally, students are asked to formulate (counter-intuitive) policy advice based on the link between structure and behavior, and to give advice related to future refinements and extensions of the model.